🏛 Phase 19: Architectural & Business Decision Log (ADR)

Document Code: DEC-LOG-001

Project: Duplicate Bin Ecosystem

Standard: Architecture Decision Records (ADR) / Business Decision Records

1. Architectural & Business Decision Ledger

Decision ID	Decision Title	Decision Date	Decision Owner	Primary Rationale & Trade-offs	Status
DEC-001	Selection of Union-Find Disjoint Sets for Duplicate Clustering	Q2 2026	Lead Architect / BA	**Chosen over Key-Based Dictionary.** Solves the transitive grouping bug ($A=B \land B=C \implies {A,B,C}$) in $O(n \alpha(n))$ near-linear time with zero group fragmentation.	**ADOPTED**
DEC-002	Adoption of 10 GB Free Deduplication Credit Baseline	Q4 2026	Product Owner / BA	**Reconciled 5 GB vs 10 GB discrepancy.** Standardizing on 10 GB in `models.py` allows creators to complete at least one full real-world	**ADOPTED**

				one full real-world	

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				cleanup workflow before being prompted to purchase credit packs.	
DEC-003	Use of SQLite WAL Mode for Persistent Hash Cache	Q2 2026	Lead Architect	**Chosen over in-memory or flat JSON files.** SQLite with Write-Ahead Logging allows concurrent thread reads and writes without database lock contention.	**ADOPTED**
DEC-004	Mandating JSON over Python `pickle` for Scan Checkpoints	Q3 2026	Security Auditor / BA	**Security Hardening Decision.** Disallows arbitrary code execution vulnerabilities associated with untrusted deserialization of Python pickle binaries.	**ADOPTED**
DEC-005	Hybrid Cloud Relay Architecture (Local Engine +	Q4 2026	Product Owner / BA	**Balances Privacy and Enterprise Governance.** File hashing executes	**ADOPTED**

	(Local Engine +			hashing executes	

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	Web Dashboard)			locally (zero file exfiltration), while high-level telemetry and commands flow through FastAPI WebSockets.	
DEC-006	LRU In-Memory Cache Cap for GUI Previews (200 Pixmaps)	Q3 2026	UI Lead / BA	**Memory Bounding Decision.** Prevents Out-Of-Memory (OOM) crashes in PySide6 when users scroll through thousands of high-resolution duplicate photo thumbnails.	**ADOPTED**